

## Experiment 10      Digital Speech Processing and Noise Reduction

### 10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

#### Program Code

```
clc; clear; close all;
```

```
Fs=8000; t=0:1/Fs:3;
```

```
x=sin(2*pi*200*t)+0.5*sin(2*pi*400*t); % Clean signal
```

```
xn=x+0.3*randn(size(x)); % Noisy signal
```

```
Y=fft(xn);
```

```
N=fft(0.3*randn(size(x)));
```

```
xe=real(ifft(max(abs(Y)-abs(N),0).*exp(1j*angle(Y))));
```

```
% Signals
```

```
figure;
```

```
subplot(311),plot(t,x),title('Clean Speech')
```

```
subplot(312),plot(t,xn),title('Noisy Speech')
```

```
subplot(313),plot(t,xe),title('Enhanced Speech')
```

```
% Spectrograms
```

```
figure;
```

```
subplot(311),spectrogram(x),title('Clean Spectrogram')
```

```
subplot(312),spectrogram(xn),title('Noisy Spectrogram')
```

```
subplot(313),spectrogram(xe),title('Enhanced Spectrogram')
```

```
% Magnitude Spectra
```

```
figure;
```

```
subplot(311),plot(abs(fft(x))),title('Clean Spectrum')
```

```
subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')
```

```
subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

## OUTPUT



